

Detailed 100-Session Teaching Plan

Agentic AI Engineering

Total Duration: 200 Hours | Total Sessions: 100 | Session Duration: 2 Hours |
Weekdays Only | Hybrid Delivery Model

Module 1: Foundations of Agentic AI			
Session No.	Date	Module Topic Title	Sub-Topics / Discussion Points
1	18-05-2026	Introduction to Agentic AI	Evolution of Agentic AI, autonomous systems, goal-driven agents
2	19-05-2026	Prompt Engineering Fundamentals	Prompt patterns, prompt chaining, prompt optimization
3	20-05-2026	Transformer Architecture	Attention mechanism, encoder-decoder models, token prediction
4	21-05-2026	Embeddings and Vector Spaces	Semantic vectors, embeddings, cosine similarity
5	22-05-2026	Reasoning Paradigms	Zero-shot reasoning, few-shot reasoning, CoT reasoning
6	25-05-2026	Agent Lifecycle	Perception, planning, reasoning, action execution
7	26-05-2026	Memory Systems	Short-term memory, long-term retrieval memory, vector memory
8	27-05-2026	Tool Invocation	Function calling, JSON outputs, API execution
9	28-05-2026	API Orchestration	REST APIs, workflow orchestration, tool integration
10	29-05-2026	Hallucination Mitigation	Grounding, retrieval validation, hallucination detection
11	01-06-2026	Optimization Strategies	Latency optimization, batching, caching
12	02-06-2026	AI Safety and RLHF	Guardrails, RLHF, policy alignment
13	03-06-2026	Prompt Injection Security	Threat vectors, mitigation techniques
14	04-06-2026	Bias and Explainability	Fairness, responsible AI, explainability
15	05-06-2026	Project related Implementation	
Module 2: LLM Ecosystem and Application Stack			
16	08-06-2026	LLM Providers	OpenAI APIs, Anthropic APIs, HuggingFace ecosystem
17	09-06-2026	API Architecture	REST APIs, endpoints, streaming responses
18	10-06-2026	Error Handling	Retry logic, rate limiting, fault handling
19	11-06-2026	Embedding Models	Semantic embeddings, vector representations
20	12-06-2026	Vector Databases	FAISS, Pinecone, Chroma architecture

21	15-06-2026	Indexing Techniques	HNSW indexing, IVF indexing
22	16-06-2026	Document Ingestion	Preprocessing pipelines, chunking strategies
23	17-06-2026	Embedding Pipelines	Indexing pipelines, retrieval pipelines
24	18-06-2026	Ranking Algorithms	Similarity search, relevance scoring
25	19-06-2026	Optimization	Latency, throughput, cost optimization
26	22-06-2026	Caching Systems	Caching strategies, embedding caching
27	23-06-2026	Async Systems	Asynchronous processing, task queues
28	24-06-2026	Retrieval Evaluation	Accuracy metrics, retrieval benchmarking
29	25-06-2026	Semantic Search	Enterprise semantic search implementation
30	26-06-2026	Project related Implementation	
Module 3: LangChain and LCEL Engineering			
31	29-06-2026	LangChain Foundations	Chains, agents, tools, retrievers
32	30-06-2026	Prompt Templates	Dynamic prompting, structured prompts
33	01-07-2026	LCEL Pipelines	Functional composition, chaining operators
34	02-07-2026	Memory Architectures	Conversation memory, vector memory
35	03-07-2026	Document Loaders	PDF loaders, CSV loaders, API ingestion
36	06-07-2026	Tool Integration	External APIs, calculators, execution tools
37	07-07-2026	QA Systems	Document question answering workflows
38	08-07-2026	Project related Implementation	
Module 4: Agent Architectures and Design Patterns			
39	09-07-2026	ReAct Framework	Reasoning-action loops, iterative execution
40	10-07-2026	Plan-and-Execute	Planner-executor workflows
41	13-07-2026	Reflexion Architecture	Self-improvement and reflection
42	14-07-2026	Stateless Agents	Sessionless execution systems
43	15-07-2026	Stateful Agents	Persistent memory systems
44	16-07-2026	Retry Strategies	Fallback systems, exponential backoff
45	17-07-2026	Guardrails	Validation pipelines, policy constraints
46	20-07-2026	Project related Implementation	
Module 5: Stateful Agent Systems with LangGraph			
47	21-07-2026	Graph Workflows	DAG workflows, cyclic workflows
48	22-07-2026	State Persistence	Checkpointing, recovery mechanisms
49	23-07-2026	Session Tracking	Context management, state tracking
50	24-07-2026	Human-in-the-loop	Approval workflows, intervention systems
51	27-07-2026	Schema-driven Outputs	Typed outputs, validation

52	28-07-2026	Workflow Orchestration	Conditional execution, branching
53	29-07-2026	Persistent Memory	Vector storage, long-term memory
54	30-07-2026	Project related Implementation	
Module 6: Retrieval-Augmented Generation (Agentic RAG)			
55	31-07-2026	RAG Foundations	Retriever-generator pipelines
56	03-08-2026	Dynamic Retrieval	Context-aware retrieval systems
57	04-08-2026	Query Rewriting	Expansion, decomposition
58	05-08-2026	Dense Retrieval	Embedding-based retrieval
59	06-08-2026	Sparse Retrieval	BM25, keyword search
60	07-08-2026	Hybrid Retrieval	Dense + sparse fusion
61	10-08-2026	LlamaIndex	Indexing workflows, query engines
62	11-08-2026	Context Optimization	Chunking, reranking
63	12-08-2026	Hallucination Detection	Grounding and verification
64	13-08-2026	Project related Implementation	
Module 7: Multi-Agent Systems and Collaborative Intelligence (Offline)			
65	14-08-2026	Planner Agents	Task planning and orchestration
66	17-08-2026	Executor Agents	Action execution systems
67	18-08-2026	Critic and Verifier Agents	Validation and evaluation
68	19-08-2026	Shared Memory Systems	Inter-agent communication
69	20-08-2026	Task Decomposition	Hierarchical planning
70	21-08-2026	CrewAI Workflows	Role-based orchestration
71	24-08-2026	Conflict Resolution	Consensus and arbitration
72	25-08-2026	Project related Implementation	
Module 8: Advanced Autonomous Agents with AutoGen (Offline)			
73	26-08-2026	AutoGen Architecture	Conversational multi-agent systems
74	27-08-2026	Dialogue Management	Turn-taking and conversations
75	28-08-2026	Code Execution Agents	Python execution environments
76	31-08-2026	Reflection Loops	Self-improvement mechanisms
77	01-09-2026	Feedback Systems	Adaptive reasoning
78	02-09-2026	Human Supervision	Override mechanisms
79	03-09-2026	Conversational Workflows	Agent-to-agent interaction
80	04-09-2026	Project related Implementation	
Module 9: AgentOps, Observability, and Evaluation (Offline)			
81	07-09-2026	Observability Pipelines	Tracing, logging, telemetry

82	08-09-2026	LangSmith	Tracing and debugging
83	09-09-2026	Langfuse	Monitoring and analytics
84	10-09-2026	Prompt Versioning	Experimentation workflows
85	11-09-2026	A/B Testing	Prompt evaluation
86	14-09-2026	Performance Metrics	Latency, cost, accuracy
87	15-09-2026	Optimization	Continuous improvement
88	16-09-2026	Project related Implementation	
Module 10: Deployment, Scaling, and Accessible AI Systems (Offline)			
89	17-09-2026	FastAPI Deployment	Routing, middleware, async APIs
90	18-09-2026	Docker Containers	Containerization workflows
91	21-09-2026	Cloud Deployment	AWS, GCP, Azure deployment
92	22-09-2026	Serverless Systems	Cloud functions and scaling
93	23-09-2026	Load Balancing	Scalability and queues
94	24-09-2026	Flowise Pipelines	No-code AI workflows
95	25-09-2026	LangFlow Systems	Visual orchestration pipelines
96	28-09-2026	Project Implementation	
97	29-09-2026	Project Implementation	
98	30-09-2026	Project Implementation	
99	01-10-2026	Project Implementation	
100	02-10-2026	Project Implementation	

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